Garaga Apparao

ch24btech11010@iith.ac.in | garaga.apparao2006@gmail.com | +91 7396401111 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.91
LAKSHYA SCHOOL , SAMALKOT MANDAL, EAST GODAVARI DIST , AP	Class XII (CENTRAL BOARD OF SECONDARY EDUCATION)	2024	91.20%
ADITYA EM HIGH SCHOOL, GANGARAJU NAGAR, EAST GODAVARI DIST , AP	Class X (ANDHRA PRADESH STATE BOARD OF SECONDARY EDUCATION)	2022	95.83%

RELEVANT COURSES

Core courses : Applied chemistry | Chemical process calculations | Thermodynamics | Transport phenomena | Heat transfer | Chemical reaction engineering | Material science for chemical engineers | Fluid mechanics | Mechanical unit operations | Mass transfer | Chemical technology | Computational fluid dynamics | Introduction to colloids and interface science

Math courses : Calculus | Elementary linear algebra | Differential equations | Numerical methods | Applied mathematics for chemical engineering

Lab courses : Applied chemistry lab | Chemistry lab | Digital fabrication | Physics lab | Heat transfer & fluid mechanics lab

SKILLS

Programing languages : C | Python | NumPy | Pandas

Simulation tools : Ansys-Lumerical | COMSOL | Solid edge | MATLAB

PROJECTS

Simulating the plasmonic properties of gold nanorods (GNRs)

- **Ansys Lumerical Finite-difference time-domain (FDTD)** , ifor solving electromagnetic equations to obtain the absorption and scattering spectrum across the range of wavelength
- Due to high **local electric field enhancements** (because of LSPR) **GOLD NANORODS** , most commonly applicable in **THERANOSTICS of a disease**
- Arranging Gold nanorods (**aspect ratio 4.1**) in different allignments (side by side, end to end, stack in 2 , stack in 4) and light is passes through one direction and run the simulation for light passing in all 3 directions
- One direction shows **LONGITUDINAL LSPR(Electric field enhancement is shifted towards high wavelength)** and another shows **TRANSVERSE LSPR(Electric field is shifted towards low wavelength)**
- Plotting of electric field and magnetic field enhancement for different number of GNR's in MATLAB

Problem Statement : A Lean and Robust Scale-Up Strategy for an Esterification Process in the Oleochemical Industry

- **Formulating** a comprehensive scale-up strategy for producing **Isopropyl Myristate** (a key emollient and transdermal drug delivery agent) from **myristic acid and isopropyl alcohol**, optimizing process yields via continuous water removal (**Le Chatelier's principle**) and stirring for avoiding bumping
- **Managing** catalyst transition and reaction kinetics from a homogeneous liquid catalyst to pilot-scale **solid Tin Oxalate**, modeling heterogeneous rate dynamics governed by external **liquid-film mass transfer and internal pore effectiveness factors** in a porous packed bed reactor

- **Conducting HAZOP** risk analyses across pilot-scale systems and **implementing** critical safety controls, mitigating **over-pressurization, vapor leaks, and toxic dust via double mechanical seals, LEL detectors, Inconel burst disks, and nitrogen-cooled filtration**

POSITIONS OF RESPONSIBILITY

Class representative | 2026- present

- Acting as the primary point of contact between student and faculty
- Resolving academic issues on behalf of peers

EXTRACURRICULAR

- An active cadet in the **National Cadet Corps (NCC**) from the school itself enhancing **communication skills , leadership skills**
- Volunteer in **Student welfare & Alumni engagement in Chemical engineering department** , working for the better BTECH student life in IITH by solving problems in the department and also bridging the gap between the Alumni of IITH chemical engineering graduate students to present batch **by being a part of organizing Alumni meet series**
- Events and multimedia Coordinator in **Department day of Chemical engineering** , IITH on 30-31 january
- A member of **Indian Institute Of Chemical Engineers(IICHE) Hyderabad Chapter**
- Actively participating in competitive hackathons and academic symposia, including **Kinexus 2026** , and **Nano Jatha**